CHAPTER 5:
ENERGY

PREREQUISITE FOR WORK

Introduction

- Allah Ta’ala states
- Power belongs wholly to Allah Ta’ala
- Allah Ta’ala has is omnipotent. He provides energy to every body and every thing.
- Energy is a meta-physical entity.
- It does not occupy space nor does it have mass, still it exist in many forms

Cont’d

- Energy can be defined as ability to do work
- Energy can not be created nor destroyed but can be transformed from one form to another
- It is only Allah Ta’ala who can creates and has the power to destroy it
- Note
- **$E= MC^2$**
- Where **E** is energy, **M** is mass and **C** is the speed of light ($3.0x\ 10^8$)

Energy resources

- Sun
- Fossil fuels
- Nuclear
- Hydro power (water)
- Wind
- Geothermal (steam)
- Biomass from wood fuel

Forms of energy

- **Chemical energy**– is the energy stored within a substance through the bonds of chemical compounds. E.g energy stored by fuels such as coal or petrol and is given inform of heat and light
- **Electrical energy**– energy possessed by moving electrons in a conductor
- **Magnetic energy**– energy due to repulsion and attraction in magnets
- **Nuclear energy–energy** due to radioactive elements e.g uranium

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- **Potential energy–** is stored energy. Is divided into two;

1. *Gravitational p. e* – is associated with the energy stored by an object because of its location above the ground (g.p.e)
2. *Elastic potential energy* is the energy stored by any object that can stretch or compress

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- **Strain energy**– energy stored in an elastic object
- **Internal energy**– energy stored in hot objects
- **Thermal energy**– energy emitted by hot objects
- **Light energy**– energy produced by glowing objects
- **Sound energy**– energy produce to vibrating objects
- **Kinetic energy (k.e)**– energy possessed by bodies in motion

Transformation and conversion

- **Energy transformation**, also termed as **energy conversion**, is the process of changing **energy** from one of its forms into another
- The **conservation of energy principle** states that energy can neither be destroyed nor created. Instead, energy just transforms from one form into another.

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- Energy can be transferred into four ways

1. By force
2. By heating
3. By radiation
4. By electrical

For instance

Lets take a rocket lifting from the ground

Chemical energy-k.e-g.p.e-thermal energy-light energy-sound energy

Energy efficiency

- In any energy conversion, the total amount of energy before and after conversion is constant
- Energy is valuable and we should not waste it
- Using more energy than necessary increases the damage we do in the environment
- Most wasted energy ends up thermal energy
- When fuels are burned they produce heat hence losing energy to the surrounding

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- Friction is problem to the moving bodies
- It can be reduced by lubrication
- Noisy machinery, loud car engines do waste energy

Making better use of energy

Also referred to as energy conservation

- Share car/ carpool
- Reduce waste
- Plant a tree
- Do not leave appliances on standby
- Replace a regular incandescent light bulb with compact fluorescent light bulb
- Cover your food while cooking
- Use washing machine or dishwasher only when they are full

Energy calculations

- Energy is a quantity that can be calculated
- Kinetic energy is the energy possessed by moving objects

Kinetic energy depends on two factors i.e **mass** and **speed** of the moving body

- The more the mass, the greater the k.e
- The more the speed, the greater the k.e

$$K.E = 1/2ms^2$$

Kinetic energy is usually measured in units of **Joules** (J); one **Joule** is equal to 1 **kg m^2 / s^2**

Example

1. A van of mass 200kg is travelling at 10m/s. calculate its kinetic energy. If its speed increases to 20m/s, by how much does kinetic energy increase
2. A car is traveling at a velocity of 10 m/s and is has a mass of 250 Kg. Compute its Kinetic energy?
3. A man is transporting a trolley of mass 6 Kg and having Kinetic energy of 40 J. Compute its Velocity with which he is running?